

Title

Spatial repellent efficacy for malaria prevention in children: results from a cluster randomized trial

Abstract

Objective: To evaluate whether spatial repellent installation reduces malaria incidence in children in endemic regions. **Setting:** Rural households in Busia County, Kenya. **Methods:** Clusters of households were randomized to spatial repellent (n=clusters receiving product) or control arms. Children aged 6 months to 10 years received malaria surveillance through 12 months. **Results:** The spatial repellent group showed 33.4% reduction in malaria infections during interim analysis, sustained at 32.7% reduction at final analysis compared to control. Clinical manifestations of malaria and parasite prevalence were significantly lower in the repellent arm. Seasonal variation was observed, with strongest protective effect during peak transmission periods. Adherence to monthly product replacement exceeded 80%. No serious adverse health events were associated with the product. **Conclusions:** Spatial repellent installation provided meaningful malaria protection in children and represents a feasible addition to vector control strategies.